

Brutal Practice Paper B40 - Heat, Comparing Quantities, Transportation in Animals & Plants, Fractions & Decimals

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A nurse in Delhi notes a child's clinical thermometer shows 102 deg F at noon and 99.5 deg F by evening; by how many Celsius degrees did the temperature fall (use 1 deg C = 1.8 deg F for a difference)?

[Heat]

- (A) Drop of about 2.5 deg F; radiation cooled the body
- (B) Drop of about 1.4 deg C; a laboratory thermometer measures body heat
- (C) Drop of 0.8 deg C; a clinical thermometer measures body heat
- (D) Drop of about 1.8 deg C; a clinical thermometer measures body heat
- (E) Drop of 2.5 deg C; a clinical thermometer measures body heat
- (F) Drop of about 1.4 deg C; a clinical thermometer measures body heat
- (G) Drop of about 4.5 deg C; conduction cooled the body

2. Aarav mixes red and white paint in the ratio 5:3, and the mixture weighs 4.8 kg. He then adds 0.6 kg of red paint. What is the new red-to-white ratio? [Comparing Quantities]

- (A) 11:6
- (B) 5:3
- (C) 13:9
- (D) 6:5
- (E) 2:1
- (F) 9:5
- (G) 7:3
- (H) 3:2

3. A doctor counts a child's pulse as 18 beats in 15 seconds; how many times does the heart beat in 5 minutes? [Transportation in Animals & Plants]

- (A) 300 beats; pulse equals heartbeat
- (B) 18 beats; pulse equals heartbeat
- (C) 72 beats; pulse equals heartbeat
- (D) 90 beats; pulse equals heartbeat
- (E) 270 beats; pulse equals heartbeat
- (F) 360 beats; pulse equals heartbeat

4. A jug holds $3\frac{3}{4}$ litres of juice; Riya pours out $\frac{2}{5}$ of it for guests. How many litres of juice remain in the jug? [Fractions & Decimals]

- (A) 1 litre
- (B) $1\frac{1}{2}$ litres
- (C) $1\frac{1}{4}$ litres
- (D) $2\frac{3}{4}$ litres
- (E) $2\frac{1}{4}$ litres
- (F) $\frac{3}{4}$ litres
- (G) $1\frac{3}{4}$ litres
- (H) $2\frac{1}{2}$ litres

5. During an afternoon at the beach the wind blows from sea toward land; if this sea breeze lasts from 11 am to 5 pm, for how many hours does it blow and why does it flow this way? [\[Heat\]](#)

- (A) 12 hours; the breeze blows all day from land to sea
- (B) 6 hours; conduction through sand pulls the wind inland
- (C) 6 hours; sea heats faster so warm air rises over the sea
- (D) 6 hours; land heats faster so warm air rises over land
- (E) 6 hours; cold air over land sinks and pushes out to sea
- (F) 5 hours; land heats faster so warm air rises over land
- (G) 7 hours; land heats faster so warm air rises over land

6. Priya scored 36 out of 50 in a test, then her teacher added 5 bonus marks to her raw score before the maximum was also raised to 55. What percent did she finally get? [\[Comparing Quantities\]](#)

- (A) 74%
- (B) 75.5%
- (C) 80%
- (D) 82%
- (E) 74.5%
- (F) 70%
- (G) 72%

7. Riya's resting heart rate is 75 beats per minute and it rises to 120 during play; how many extra beats occur over 6 minutes of play? [\[Transportation in Animals & Plants\]](#)

- (A) 45 extra beats; exercise raises heart rate
- (B) 720 extra beats; exercise raises heart rate
- (C) 195 extra beats; exercise raises heart rate
- (D) 225 extra beats; exercise raises heart rate
- (E) 270 extra beats; exercise raises heart rate
- (F) 450 extra beats; exercise raises heart rate
- (G) 270 extra beats; rest lowers heart rate

8. Sunlight crosses a glass panel and its brightness becomes 0.6 of the original; through a second panel it again becomes 0.6. What fraction of the starting brightness remains? [\[Fractions & Decimals\]](#)

- (A) $\frac{6}{25}$
- (B) $\frac{18}{25}$
- (C) $\frac{3}{5}$
- (D) $\frac{9}{25}$
- (E) $\frac{12}{25}$
- (F) $\frac{6}{10}$
- (G) $\frac{1}{5}$
- (H) $\frac{36}{100}$

9. A steel railway track of length 200 m is laid in winter; in summer it expands by 0.06% of its length, so what gap must the workers leave and what property causes it? [\[Heat\]](#)

- (A) 1.2 m gap; solids expand on heating
- (B) 0.12 m gap; solids contract on heating
- (C) 0.12 cm gap; radiation lengthens the steel
- (D) 0.06 m gap; metals are good insulators
- (E) 12 m gap; solids expand on heating
- (F) 0.24 m gap; solids expand on heating
- (G) 0.12 m gap; solids expand on heating

10. A bicycle's price is first increased by 20% and then the new price is decreased by 20%. If the original price was Rs 5000, what is the final price? [\[Comparing Quantities\]](#)

- (A) Rs 5000
- (B) Rs 6000
- (C) Rs 5200
- (D) Rs 4800
- (E) Rs 4000
- (F) Rs 4900

11. Blood is about 55% plasma and the rest cells; in 4 litres of blood, how much volume is cells, in millilitres? [\[Transportation in Animals & Plants\]](#)

- (A) 1600 mL; cells are 40% of blood
- (B) 1800 mL; plasma is 45% of blood
- (C) 2450 mL; cells are 45% of blood
- (D) 1800 mL; cells are 45% of blood
- (E) 2000 mL; cells are 50% of blood
- (F) 180 mL; cells are 45% of blood
- (G) 2200 mL; cells are 55% of blood

12. Tiny screws each weigh 0.025 kg; a box has 2.4 kg of them. Each screw is then cut in half by weight. How many half-screws come from the box? [\[Fractions & Decimals\]](#)

- (A) 192
- (B) 120
- (C) 96
- (D) 960
- (E) 144
- (F) 19.2
- (G) 48

13. Reema mixes 3 litres of water at 80 deg C with 2 litres of water at 30 deg C in an insulated pot; what is the final temperature using a weighted average, and which transfer evens it out? [\[Heat\]](#)

- (A) 60 deg F; conduction inside the water mixes the heat
- (B) 55 deg C; conduction inside the water mixes the heat
- (C) 60 deg C; conduction inside the water mixes the heat
- (D) 65 deg C; conduction inside the water mixes the heat
- (E) 60 deg C; conduction makes the water hotter than both
- (F) 60 deg C; radiation inside the water mixes the heat

14. Tina buys 15 pens for Rs 180 and sells all of them so as to gain the cost of 3 pens. Find her profit percent. [\[Comparing Quantities\]](#)

- (A) 22%
- (B) 16.67%
- (C) 18%
- (D) 30%
- (E) 20%
- (F) 25%
- (G) 15%

15. A leafy plant loses 6 mL of water by transpiration each hour during 8 daylight hours; if roots absorb 60 mL that day, how much water is kept by the plant? [\[Transportation in Animals & Plants\]](#)

- (A) 0 mL; most absorbed water is transpired
- (B) 18 mL; most absorbed water is transpired
- (C) 60 mL; all absorbed water is transpired
- (D) 48 mL; most absorbed water is transpired
- (E) 12 mL; most absorbed water is transpired
- (F) 54 mL; most absorbed water is transpired
- (G) 6 mL; most absorbed water is transpired

16. A recipe needs $\frac{2}{3}$ cup of flour per batch. Devi bakes $4\frac{1}{2}$ batches but spills $\frac{1}{6}$ cup. How much flour did she actually use? [\[Fractions & Decimals\]](#)

- (A) $3\frac{1}{6}$ cups
- (B) $3\frac{1}{2}$ cups
- (C) $2\frac{1}{2}$ cups
- (D) $2\frac{1}{6}$ cups
- (E) $2\frac{2}{3}$ cups
- (F) $3\frac{1}{3}$ cups
- (G) $2\frac{5}{6}$ cups
- (H) 3 cups

17. Two identical cups, one black and one white, sit in sunlight; after 10 minutes the black cup rose 12 deg C and the white rose 4 deg C, so how many times faster did the black one heat and why? [\[Heat\]](#)

- (A) 12 times faster; dark surfaces absorb more radiation
- (B) 8 times faster; dark surfaces absorb more radiation
- (C) 2 times faster; dark surfaces absorb more radiation
- (D) 1.5 times faster; dark surfaces absorb more radiation
- (E) 3 times faster; dark surfaces absorb more radiation
- (F) 3 times faster; light surfaces absorb more radiation

18. Rahul deposits Rs 8000 at 6% simple interest per year. After 2 years he withdraws everything and reinvests the full amount at 5% for 1 more year. What total interest has he earned overall? [\[Comparing Quantities\]](#)

- (A) Rs 1360
- (B) Rs 1408.40
- (C) Rs 1280
- (D) Rs 1408
- (E) Rs 960
- (F) Rs 1448

- 19.** Counting in a smear, every 600 red blood cells appear with 1 white blood cell; in a sample showing 9 white cells, how many red cells are present? [\[Transportation in Animals & Plants\]](#)
- (A) 4800 red cells; RBCs far outnumber WBCs
 - (B) 609 red cells; RBCs far outnumber WBCs
 - (C) 60 red cells; RBCs far outnumber WBCs
 - (D) 5400 red cells; RBCs far outnumber WBCs
 - (E) 600 red cells; RBCs far outnumber WBCs
 - (F) 5409 red cells; RBCs far outnumber WBCs
 - (G) 5400 white cells; WBCs far outnumber RBCs
- 20.** Compare carefully: which is larger, the product $\frac{3}{4}$ times $\frac{5}{6}$, or the product $\frac{2}{3}$ times $\frac{9}{10}$? [\[Fractions & Decimals\]](#)
- (A) Cannot be compared
 - (B) They are equal
 - (C) $\frac{2}{3}$ times $\frac{9}{10}$
 - (D) $\frac{3}{5}$
 - (E) $\frac{3}{4}$ times $\frac{5}{6}$
 - (F) $\frac{5}{8}$
- 21.** Grandma's kitchen has a metal spoon and a wooden spoon in the same hot dal; after 1 minute the metal feels much hotter to touch - by what mechanism, and which spoon is the better insulator? [\[Heat\]](#)
- (A) Conduction; the wooden spoon is the better conductor
 - (B) Conduction; the wooden spoon is the better insulator
 - (C) Radiation; the wooden spoon is the better insulator
 - (D) Conduction; the metal spoon is the better insulator
 - (E) Radiation; the metal spoon is the better insulator
 - (F) Convection; the wooden spoon is the better insulator
 - (G) Conduction; both spoons are equal insulators
- 22.** On a shirt marked Rs 1200, a store gives 25% discount and then charges 8% tax on the discounted price. What does the customer pay? [\[Comparing Quantities\]](#)
- (A) Rs 996
 - (B) Rs 972
 - (C) Rs 900
 - (D) Rs 1011
 - (E) Rs 1000
 - (F) Rs 972.50
 - (G) Rs 1296
- 23.** With every single beat the heart ejects about 70 mL of blood; at 72 beats per minute, how many litres does it pump in 10 minutes? [\[Transportation in Animals & Plants\]](#)
- (A) 5.04 mL; stroke volume times rate gives output
 - (B) 504 litres; stroke volume times rate gives output
 - (C) 5.04 litres; stroke volume times rate gives output
 - (D) 5040 litres; stroke volume times rate gives output
 - (E) 50.4 litres; one beat pumps the whole body's blood
 - (F) 50.4 litres; stroke volume times rate gives output
 - (G) 36 litres; stroke volume times rate gives output

24. Walking briskly, Omar covers $\frac{4}{5}$ km every $\frac{1}{4}$ hour. At this rate, how far does he walk in $1\frac{1}{2}$ hours?

[Fractions & Decimals]

- (A) $5\frac{1}{5}$ km
- (B) 3 km
- (C) $4\frac{4}{5}$ km
- (D) $2\frac{2}{5}$ km
- (E) $1\frac{1}{5}$ km
- (F) 6 km
- (G) $3\frac{1}{5}$ km

25. A laboratory thermometer ranges from -10°C to 110°C ; if each small mark is 1°C and the reading sits 8 marks above the 25°C line, what is the temperature and what tool is this? [Heat]

- (A) 35°C ; a laboratory thermometer reads a wide range
- (B) 33°C ; a clinical thermometer reads body heat only
- (C) 33°C ; a clinical thermometer reads a wide range
- (D) 8°C ; a laboratory thermometer reads a wide range
- (E) 17°C ; a laboratory thermometer reads a wide range
- (F) 33°C ; a laboratory thermometer reads a wide range
- (G) 33°F ; a laboratory thermometer reads a wide range
- (H) 32°C ; a laboratory thermometer reads a wide range

26. Out of 40 students, the ratio of those who passed to those who failed is 7:1. If 35% of the passers scored above 90, how many passers scored above 90? [Comparing Quantities]

- (A) 14
- (B) 12.25, rounded to 12
- (C) 12.25
- (D) 7
- (E) 10
- (F) 13
- (G) 11

27. Each gram of haemoglobin can carry about 1.3 mL of oxygen; if blood holds 150 g of haemoglobin, how much oxygen can it carry? [Transportation in Animals & Plants]

- (A) 1950 mL oxygen; haemoglobin binds oxygen
- (B) 195 mL oxygen; plasma binds oxygen
- (C) 151.3 mL oxygen; haemoglobin binds oxygen
- (D) 115 mL oxygen; haemoglobin binds oxygen
- (E) 150 mL oxygen; haemoglobin binds oxygen
- (F) 195 mL oxygen; haemoglobin binds oxygen
- (G) 19.5 mL oxygen; haemoglobin binds oxygen

28. Express 0.75 times 0.8 as a fraction, then state that fraction reduced to its simplest form. [[Fractions & Decimals](#)]

- (A) $\frac{5}{3}$
- (B) $\frac{60}{100}$
- (C) $\frac{6}{10}$
- (D) $\frac{12}{20}$
- (E) $\frac{2}{5}$
- (F) 0.6
- (G) $\frac{3}{5}$
- (H) $\frac{3}{10}$

29. A thermos keeps tea at 70 deg C and loses only 2 deg C every hour, while an open cup loses 10 deg C each hour; after 3 hours how much warmer is the thermos tea and what design stops the loss? [[Heat](#)]

- (A) 8 deg C warmer; a vacuum gap stops conduction and convection
- (B) 16 deg C warmer; a vacuum gap stops conduction and convection
- (C) 24 deg C warmer; a vacuum gap stops conduction and convection
- (D) 24 deg C warmer; shiny walls increase radiation loss
- (E) 6 deg C warmer; a vacuum gap stops conduction and convection
- (F) 30 deg C warmer; a vacuum gap stops conduction and convection
- (G) 24 deg C warmer; a vacuum gap speeds conduction and convection

30. A trader buys oranges at Rs 40 per dozen and sells them at Rs 5 each. Find his profit and profit percent. [[Comparing Quantities](#)]

- (A) Rs 60 profit; a 60% profit
- (B) Rs 60 profit; a 150% profit
- (C) Rs 20 profit; a 20% profit
- (D) Rs 20 profit; a 50% profit
- (E) Rs 20 profit; a 25% profit
- (F) Rs 20 profit; a 33.3% profit
- (G) Rs 20 loss; a 50% loss

31. Healthy kidneys filter blood and form about 1.5 litres of urine daily; how many millilitres of urine form in 4 days? [[Transportation in Animals & Plants](#)]

- (A) 60000 mL; kidneys remove waste as urine
- (B) 1500 mL; kidneys remove waste as urine
- (C) 6000 mL; kidneys remove waste as sweat
- (D) 4500 mL; kidneys remove waste as urine
- (E) 6 mL; kidneys remove waste as urine
- (F) 6000 mL; kidneys remove waste as urine

32. A water tank is $\frac{7}{8}$ full. After 0.25 of the full tank is drained, what fraction of the tank still has water?

[[Fractions & Decimals](#)]

- (A) $\frac{1}{2}$
- (B) $\frac{5}{8}$
- (C) $\frac{7}{16}$
- (D) $\frac{5}{16}$
- (E) 0.625
- (F) $\frac{3}{4}$
- (G) $\frac{3}{8}$

33. At night the wind reverses and blows from land to sea between 9 pm and 3 am; how long is this land breeze and why does it now flow toward the sea? [\[Heat\]](#)

- (A) 3 hours; sea stays warmer so air rises over the sea
- (B) 8 hours; sea stays warmer so air rises over the sea
- (C) 6 hours; sea stays warmer so air rises over the sea
- (D) 6 hours; land stays warmer so air rises over the land
- (E) 12 hours; the breeze blows toward land all night
- (F) 6 hours; conduction through water pulls air seaward

34. In a class, boys and girls are in the ratio 3:2. If the number of girls increases by 25% while boys stay the same, what is the new ratio of boys to girls? [\[Comparing Quantities\]](#)

- (A) 12:10
- (B) 5:6
- (C) 9:5
- (D) 3:5
- (E) 3:2
- (F) 6:5
- (G) 6:4

35. Through a single stoma a leaf loses 0.5 microlitre of water hourly; a leaf with 2000 stomata transpires for 3 hours, losing how much water? [\[Transportation in Animals & Plants\]](#)

- (A) 1500 microlitres; stomata release water vapour
- (B) 3000 microlitres; root hairs release water vapour
- (C) 6000 microlitres; stomata release water vapour
- (D) 3000 microlitres; stomata release water vapour
- (E) 2000 microlitres; stomata release water vapour
- (F) 300 microlitres; stomata release water vapour
- (G) 1000 microlitres; stomata release water vapour

36. Pencils cost Rs 3.50 each. Meena buys $\frac{2}{3}$ of a dozen and pays with a Rs 50 note. How much change does she get? [\[Fractions & Decimals\]](#)

- (A) Rs 22
- (B) Rs 14
- (C) Rs 26
- (D) Rs 36
- (E) Rs 22.50
- (F) Rs 18
- (G) Rs 24
- (H) Rs 28

37. A child's normal body temperature is 37 deg C; convert this to Fahrenheit using $F = (C \times 9/5) + 32$, and say which thermometer has a kink to hold this reading. [\[Heat\]](#)

- (A) 98.6 deg C; a clinical thermometer has a kink
- (B) 66.6 deg F; a clinical thermometer has a kink
- (C) 69 deg F; a clinical thermometer has a kink
- (D) 98.6 deg F; a clinical thermometer has a kink
- (E) 98.6 deg F; a laboratory thermometer has a kink
- (F) 98.6 deg F; a clinical thermometer has no kink
- (G) 37 deg F; a clinical thermometer has a kink
- (H) 100.4 deg F; a clinical thermometer has a kink

38. Sameer sells a watch for Rs 1980 at a 10% loss. To make a 5% profit instead, at what price should he have sold it? [\[Comparing Quantities\]](#)

- (A) Rs 2090
- (B) Rs 2200
- (C) Rs 2100
- (D) Rs 2376
- (E) Rs 2178
- (F) Rs 2310
- (G) Rs 2079

39. An athlete's heart rate is 60 bpm at rest and 150 bpm while running; if she runs 4 minutes then rests 6 minutes, how many total heartbeats occur in those 10 minutes? [\[Transportation in Animals & Plants\]](#)

- (A) 960 beats; rate stays the same always
- (B) 1050 beats; rate changes with activity
- (C) 600 beats; rate changes with activity
- (D) 840 beats; rate changes with activity
- (E) 1500 beats; rate changes with activity
- (F) 960 beats; rate changes with activity
- (G) 900 beats; rate changes with activity

40. Ribbon 9.6 metres long is cut into pieces each 0.8 metre; then every piece is folded into 4 equal parts. How many folded parts result? [\[Fractions & Decimals\]](#)

- (A) 1.2
- (B) 3
- (C) 12
- (D) 9.6
- (E) 48
- (F) 24
- (G) 40
- (H) 16

41. A pot of water on a flame shows the bottom layer heating first, then the hot water rises and cool water sinks in a loop; name this transfer and the average rise if it climbs 5 deg C, 7 deg C and 9 deg C in three minutes. [\[Heat\]](#)

- (A) Convection; average rise 5 deg C per minute
- (B) Convection; average rise 7 deg C per minute
- (C) Convection; average rise 21 deg C per minute
- (D) Radiation; average rise 7 deg C per minute
- (E) Conduction; average rise 21 deg C per minute
- (F) Convection; average rise 8 deg C per minute

42. Water makes up 75% of a 60 kg mixture. After some water evaporates, water is now 60% of the mixture. How much water evaporated? [\[Comparing Quantities\]](#)

- (A) 7.5 kg
- (B) 22.5 kg
- (C) 9 kg
- (D) 10 kg
- (E) 18 kg
- (F) 12 kg
- (G) 15 kg

43. In an emergency a patient needs 25% of an adult's 5-litre blood volume replaced; how many millilitres of blood must be transfused? [\[Transportation in Animals & Plants\]](#)

- (A) 1250 mL; blood volume is about 5 litres
- (B) 1500 mL; blood volume is about 5 litres
- (C) 1250 mL; blood volume is about 1 litre
- (D) 2500 mL; blood volume is about 5 litres
- (E) 125 mL; blood volume is about 5 litres
- (F) 1000 mL; blood volume is about 5 litres
- (G) 12500 mL; blood volume is about 5 litres

44. Three-fifths of a class of 40 students like cricket; half of those who like cricket also like football. How many like both? [\[Fractions & Decimals\]](#)

- (A) 24
- (B) 8
- (C) 20
- (D) 16
- (E) 6
- (F) 12
- (G) 15

45. A clinical thermometer can only be read up to 42 deg C; a patient's reading rose from 38.5 deg C to 40.6 deg C over the illness - what was the rise, and why can this thermometer not measure boiling water?

[Heat]

- (A) 2.1 deg C rise; its range is wide enough for boiling water
- (B) 2.1 deg C rise; it lacks a kink for boiling water
- (C) 1.1 deg C rise; its range is too small for boiling water
- (D) 1.9 deg C rise; its range is too small for boiling water
- (E) 2.1 deg F rise; its range is too small for boiling water
- (F) 2.0 deg C rise; its range is too small for boiling water
- (G) 2.1 deg C rise; its range is too small for boiling water

46. Two numbers are in ratio 4:7 and their sum is 143. By what percent is the larger number more than the smaller? [Comparing Quantities]

- (A) 43.75%
- (B) 60%
- (C) 57%
- (D) 70%
- (E) 75%
- (F) 175%
- (G) 75.5%

47. A breathing-rate study shows 20 breaths per minute and a heartbeat of 80 per minute; over 3 minutes, how many more heartbeats than breaths occur? [Transportation in Animals & Plants]

- (A) 240 more beats; heart beats faster than we breathe
- (B) 180 more breaths; we breathe faster than the heart beats
- (C) 60 more beats; heart beats faster than we breathe
- (D) 100 more beats; heart beats faster than we breathe
- (E) 300 more beats; heart beats faster than we breathe
- (F) 180 more beats; heart beats faster than we breathe
- (G) 60 more breaths; we breathe faster than the heart beats

48. A car uses 1 1/4 litres of fuel per 10 km. For a 36 km trip, how many litres of fuel are needed? [Fractions & Decimals]

- (A) 4 4/5 litres
- (B) 3 1/2 litres
- (C) 4 1/4 litres
- (D) 5 litres
- (E) 45 litres
- (F) 4 1/2 litres
- (G) 4 litres

49. An iron ball just fits through a ring at 20 deg C; when heated it no longer passes through, but after cooling to 5 deg C it slips through easily - explain using expansion and find the total temperature swing.

[Heat]

- (A) 20 deg C swing; metals expand on heating and shrink on cooling
- (B) 25 deg C swing; metals expand on heating and shrink on cooling
- (C) 35 deg C swing; metals expand on heating and shrink on cooling
- (D) 15 deg C swing; only liquids expand on heating
- (E) 5 deg C swing; metals expand on heating and shrink on cooling
- (F) 15 deg C swing; metals expand on heating and shrink on cooling
- (G) 15 deg C swing; metals shrink on heating and expand on cooling

50. A shopkeeper marks goods 40% above cost, then offers a 25% discount on the marked price. What is his profit percent? [Comparing Quantities]

- (A) 65%
- (B) 10%
- (C) 5%
- (D) 4%
- (E) 15%
- (F) 12.5%
- (G) 20%

51. Phloem carries food made in leaves; if a leaf makes 40 mg of sugar per hour and sends 75% to the roots, how much sugar reaches the roots in 4 hours? [Transportation in Animals & Plants]

- (A) 40 mg; phloem transports food downward
- (B) 120 mg; xylem transports food downward
- (C) 480 mg; phloem transports food downward
- (D) 120 mg; phloem transports food downward
- (E) 160 mg; phloem transports food downward
- (F) 30 mg; phloem transports food downward

52. Multiply 2.5 by 0.04, then divide the result by 0.2. What decimal do you finally obtain? [Fractions & Decimals]

- (A) 0.25
- (B) 5
- (C) 0.5
- (D) 2.5
- (E) 0.05
- (F) 0.005

53. A solar water heater on a roof raises 50 litres of water by 20 deg C using sunlight; if a black-painted tank absorbs heat twice as fast as a silver one, how long does black take if silver needs 60 minutes, and why? [Heat]

- (A) 120 minutes; black absorbs radiation faster than silver
- (B) 60 minutes; black absorbs radiation faster than silver
- (C) 45 minutes; black absorbs radiation faster than silver
- (D) 30 minutes; black reflects radiation faster than silver
- (E) 30 minutes; conduction from the roof heats the tank
- (F) 30 minutes; silver absorbs radiation faster than black
- (G) 30 minutes; black absorbs radiation faster than silver

- 54.** A car travels 240 km on 16 litres of fuel. At the same rate, how many litres are needed for a trip of 375 km? [\[Comparing Quantities\]](#)
- (A) 26 litres
 - (B) 24 litres
 - (C) 23 litres
 - (D) 20 litres
 - (E) 25 litres
 - (F) 22.5 litres
 - (G) 30 litres
- 55.** A dialysis machine cleans a patient's blood for 4 hours at 250 mL of blood processed per minute; how many litres of blood does it process? [\[Transportation in Animals & Plants\]](#)
- (A) 60 litres; dialysis replaces kidney filtering
 - (B) 60 litres; dialysis replaces heart pumping
 - (C) 1 litre; dialysis replaces kidney filtering
 - (D) 60 mL; dialysis replaces kidney filtering
 - (E) 6 litres; dialysis replaces kidney filtering
 - (F) 15 litres; dialysis replaces kidney filtering
- 56.** Half of a $2\frac{2}{5}$ metre rope is used, then $\frac{3}{4}$ of what is left gets used. How much rope remains unused? [\[Fractions & Decimals\]](#)
- (A) $\frac{6}{10}$ metre
 - (B) $1\frac{1}{5}$ metre
 - (C) $\frac{1}{2}$ metre
 - (D) $\frac{2}{5}$ metre
 - (E) $\frac{3}{5}$ metre
 - (F) $\frac{3}{10}$ metre
 - (G) $\frac{9}{10}$ metre
 - (H) $\frac{1}{4}$ metre
- 57.** On a hot day a person wears a dark shirt and feels hotter than a friend in white; if dark absorbs 80% of sunlight and white only 20%, how many times more heat does the dark shirt take in? [\[Heat\]](#)
- (A) 4 times more; dark colours absorb more radiation
 - (B) 3 times more; dark colours absorb more radiation
 - (C) 60 times more; dark colours absorb more radiation
 - (D) 4 times more; conduction warms the dark shirt
 - (E) 2 times more; dark colours absorb more radiation
 - (F) 4 times more; light colours absorb more radiation
 - (G) 4 times more; dark colours reflect more radiation
- 58.** Megha lent Rs 12000, part at 8% and the rest at 10% simple interest for 1 year, earning Rs 1080 total interest. How much was lent at 8%? [\[Comparing Quantities\]](#)
- (A) Rs 8000
 - (B) Rs 6500
 - (C) Rs 5500
 - (D) Rs 5000
 - (E) Rs 7000
 - (F) Rs 6000
 - (G) Rs 4000

- 59.** Water rises in xylem at 15 cm per minute; a tree stem is 3 metres tall, so how long does water take to reach the top, in minutes? [\[Transportation in Animals & Plants\]](#)
- (A) 45 minutes; xylem carries water upward
 - (B) 5 minutes; xylem carries water upward
 - (C) 20 minutes; phloem carries water upward
 - (D) 20 minutes; xylem carries water upward
 - (E) 30 minutes; xylem carries water upward
 - (F) 200 minutes; xylem carries water upward
 - (G) 2 minutes; xylem carries water upward
- 60.** Which decimal lies exactly between the fractions $\frac{3}{8}$ and $\frac{1}{2}$ on the number line? [\[Fractions & Decimals\]](#)
- (A) 0.375
 - (B) 0.4375
 - (C) 0.43
 - (D) 0.45
 - (E) 0.4
 - (F) 0.5
- 61.** A double-walled thermos flask has a shiny silver inner surface; this mirror-like coating mainly cuts down which kind of heat loss, and if it saves 25% of a 40 deg C drop, how much drop is saved? [\[Heat\]](#)
- (A) 15 deg C saved; shiny surfaces cut radiation loss
 - (B) 10 deg C saved; shiny surfaces cut radiation loss
 - (C) 10 deg C saved; shiny surfaces cut conduction loss
 - (D) 4 deg C saved; shiny surfaces cut radiation loss
 - (E) 25 deg C saved; shiny surfaces cut radiation loss
 - (F) 30 deg C saved; shiny surfaces cut radiation loss
- 62.** A jacket costs Rs 2500. Its price rises 12% in winter, then drops 12% in spring. Compare the spring price with the original. [\[Comparing Quantities\]](#)
- (A) Rs 2200; a 12% decrease
 - (B) Rs 2500; no change
 - (C) Rs 2536; a 1.44% increase
 - (D) Rs 2440; a 2.4% decrease
 - (E) Rs 2464; a 1.44% increase
 - (F) Rs 2464; a 1.44% decrease
- 63.** A blood drop has platelets that help clotting; if 1 microlitre holds 250000 platelets, how many platelets are in 6 microlitres? [\[Transportation in Animals & Plants\]](#)
- (A) 15000000 platelets; platelets help blood clot
 - (B) 1500000 platelets; platelets help blood clot
 - (C) 150000 platelets; platelets help blood clot
 - (D) 1500000 platelets; platelets carry oxygen
 - (E) 250000 platelets; platelets help blood clot
 - (F) 1250000 platelets; platelets help blood clot

- 64.** A shopkeeper sells $\frac{3}{4}$ kg of sugar to one buyer and 1.5 kg to another from a 5 kg sack. What fraction of the sack is left? [\[Fractions & Decimals\]](#)
- (A) $\frac{9}{20}$
 - (B) $\frac{7}{20}$
 - (C) $\frac{3}{5}$
 - (D) $\frac{2}{5}$
 - (E) $\frac{1}{2}$
 - (F) $\frac{13}{20}$
 - (G) $\frac{3}{4}$
 - (H) $\frac{11}{20}$
- 65.** Heat flows only from a hotter body to a cooler one; if a 90 deg C iron rod touches a 30 deg C block and they meet halfway at equal temperature, what is that temperature and which way did heat flow? [\[Heat\]](#)
- (A) 60 deg F; heat flowed from the hot rod to the cool block
 - (B) 45 deg C; heat flowed from the hot rod to the cool block
 - (C) 90 deg C; heat flowed from the hot rod to the cool block
 - (D) 60 deg C; heat flowed from the hot rod to the cool block
 - (E) 60 deg C; heat flowed from the cool block to the hot rod
 - (F) 50 deg C; heat flowed from the hot rod to the cool block
 - (G) 120 deg C; heat flowed from the hot rod to the cool block
- 66.** A recipe for 6 people needs 450 g flour. Neha cooks for 10 people but has only 600 g flour. By what percent does she fall short of the flour needed? [\[Comparing Quantities\]](#)
- (A) 15% short
 - (B) 20% short
 - (C) 25% short
 - (D) 10% short
 - (E) 16.67% short
 - (F) 33.3% short
- 67.** Comparing two pupils, Sam's pulse is 16 beats in 12 seconds and Tina's is 21 beats in 15 seconds; whose heart beats faster per minute and by how much? [\[Transportation in Animals & Plants\]](#)
- (A) Sam by 4 bpm; pulse rate equals heart rate
 - (B) Tina by 5 bpm; pulse rate equals heart rate
 - (C) Tina by 4 bpm; pulse rate is double heart rate
 - (D) Tina by 84 bpm; pulse rate equals heart rate
 - (E) Sam by 5 bpm; pulse rate equals heart rate
 - (F) Tina by 4 bpm; pulse rate equals heart rate
 - (G) Equal; pulse rate equals heart rate

68. Reciprocal trap: divide $\frac{5}{6}$ by $2\frac{1}{2}$, then multiply the quotient by $\frac{3}{4}$. What is the result? [\[Fractions & Decimals\]](#)

- (A) $\frac{1}{2}$
- (B) $\frac{1}{4}$
- (C) $\frac{5}{16}$
- (D) $\frac{2}{3}$
- (E) $\frac{25}{12}$
- (F) $\frac{5}{8}$
- (G) $\frac{1}{3}$
- (H) $\frac{3}{8}$

69. A laboratory thermometer is read with the eye level to the liquid; if a student wrongly reads 8 deg C too high and the true value is 46 deg C, what wrong number did they get and what is the correct way to read it?

[\[Heat\]](#)

- (A) 8 deg C wrong; keep the eye level with the liquid
- (B) 54 deg C wrong; tilt the thermometer while reading
- (C) 38 deg C wrong; keep the eye level with the liquid
- (D) 54 deg F wrong; keep the eye level with the liquid
- (E) 52 deg C wrong; keep the eye level with the liquid
- (F) 46 deg C wrong; keep the eye level with the liquid
- (G) 54 deg C wrong; keep the eye level with the liquid
- (H) 54 deg C wrong; keep the eye above the liquid

70. An item is sold for Rs 575 at a 15% profit. If instead it is sold for Rs 450, find the profit or loss percent.

[\[Comparing Quantities\]](#)

- (A) a 9% loss
- (B) a 10% profit
- (C) a 15% loss
- (D) a 10% loss
- (E) a 21.7% loss
- (F) a 12% loss
- (G) a 11% loss

71. The body has roughly 5 litres of blood and the heart pumps about 5 litres each minute; how many minutes does it take to pump blood equal to 35 litres? [\[Transportation in Animals & Plants\]](#)

- (A) 7 minutes; blood is pumped only once a day
- (B) 5 minutes; whole blood volume recirculates often
- (C) 35 minutes; whole blood volume recirculates often
- (D) 1 minute; whole blood volume recirculates often
- (E) 7 minutes; whole blood volume recirculates often
- (F) 175 minutes; whole blood volume recirculates often
- (G) 8.75 minutes; whole blood volume recirculates often

72. Gold wire weighs 0.125 kg per metre. A jeweller has 3.6 kg of wire and uses $\frac{2}{3}$ of it. How many metres did he use? [\[Fractions & Decimals\]](#)

- (A) 21.6
- (B) 9.6
- (C) 19.2
- (D) 0.3
- (E) 14.4
- (F) 8

73. Sand heats and cools faster than seawater; if at noon the sand is 45 deg C and the sea is 27 deg C, find the difference and explain how this gap creates the daytime breeze. [\[Heat\]](#)

- (A) 18 deg C gap; warm air rises over land, drawing a sea breeze
- (B) 18 deg F gap; warm air rises over land, drawing a sea breeze
- (C) 12 deg C gap; warm air rises over land, drawing a sea breeze
- (D) 72 deg C gap; warm air rises over land, drawing a sea breeze
- (E) 18 deg C gap; warm air rises over the sea, drawing a land breeze
- (F) 18 deg C gap; cool air sinks over land, drawing a land breeze
- (G) 27 deg C gap; warm air rises over land, drawing a sea breeze

74. Of a Rs 60000 salary, 30% goes to rent and 25% of the remainder goes to food. How much is left after rent and food? [\[Comparing Quantities\]](#)

- (A) Rs 27000
- (B) Rs 28500
- (C) Rs 31500
- (D) Rs 31000
- (E) Rs 32500
- (F) Rs 30000
- (G) Rs 33000

75. On a hot day a person sweats out 1200 mL of water and drinks back 800 mL; ignoring the salts lost, how many millilitres of water are still lost overall? [\[Transportation in Animals & Plants\]](#)

- (A) 400 mL; sweat removes water and salts
- (B) 840 mL; sweat removes water and salts
- (C) 1200 mL; sweat removes water and salts
- (D) 2000 mL; sweat removes water and salts
- (E) 400 mL; sweat removes only salts
- (F) 560 mL; sweat removes water and salts
- (G) 360 mL; sweat removes water and salts

76. Rohan reads $\frac{2}{5}$ of a 250-page book on Monday and $\frac{1}{3}$ of the remainder on Tuesday. How many pages are still unread? [\[Fractions & Decimals\]](#)

- (A) 120
- (B) 150
- (C) 110
- (D) 100
- (E) 50
- (F) 90
- (G) 75
- (H) 60

77. Three rods of copper, plastic and glass each touch boiling water; copper passes heat fastest and plastic slowest, so rank them as conductor or insulator and find the time for plastic if copper takes 4 s and plastic is 9 times slower. [\[Heat\]](#)

- (A) 36 seconds; copper is the best conductor, plastic the best insulator
- (B) 36 seconds; plastic is the best conductor, copper the best insulator
- (C) 32 seconds; copper is the best conductor, plastic the best insulator
- (D) 13 seconds; copper is the best conductor, plastic the best insulator
- (E) 36 minutes; copper is the best conductor, plastic the best insulator
- (F) 45 seconds; copper is the best conductor, plastic the best insulator
- (G) 36 seconds; copper and plastic are equal conductors
- (H) 0.44 seconds; copper is the best conductor, plastic the best insulator

78. A sum doubles itself in 8 years at simple interest. What is the annual rate of interest? [\[Comparing Quantities\]](#)

- (A) 8%
- (B) 12%
- (C) 12.5%
- (D) 10%
- (E) 6.25%
- (F) 25%

79. Arteries carry blood away from the heart with thick walls; if an artery wall is 3 times as thick as a vein wall measuring 0.4 mm, how thick is the artery wall? [\[Transportation in Animals & Plants\]](#)

- (A) 3.4 mm; arteries have thicker walls than veins
- (B) 1.2 mm; arteries have thicker walls than veins
- (C) 1.6 mm; arteries have thicker walls than veins
- (D) 12 mm; arteries have thicker walls than veins
- (E) 0.4 mm; arteries have thicker walls than veins
- (F) 0.12 mm; arteries have thicker walls than veins
- (G) 1.2 mm; veins have thicker walls than arteries

80. Convert $1\frac{3}{8}$ into a decimal, multiply it by 0.4, and write the answer as a decimal. [\[Fractions & Decimals\]](#)

- (A) 0.625
- (B) 0.5
- (C) 1.775
- (D) 0.44
- (E) 0.55
- (F) 0.6

81. A fever chart shows readings of 38 deg C, 39 deg C, 40 deg C and 39 deg C over four checks; find the average fever and state which thermometer with a kink was used to record each value. [\[Heat\]](#)

- (A) 39 deg C average; a clinical thermometer with a kink was used
- (B) 40 deg C average; a clinical thermometer with a kink was used
- (C) 38.5 deg C average; a clinical thermometer with a kink was used
- (D) 39.5 deg C average; a clinical thermometer with a kink was used
- (E) 39 deg C average; a clinical thermometer with no kink was used
- (F) 39 deg F average; a clinical thermometer with a kink was used

- 82.** The price of sugar rises from Rs 40 to Rs 50 per kg. By what percent must a family cut its consumption to keep the same total spending? [\[Comparing Quantities\]](#)
- (A) 12.5%
 - (B) 10%
 - (C) 25%
 - (D) 22%
 - (E) 15%
 - (F) 20%
 - (G) 20.5%
- 83.** The right side of the heart sends blood to lungs and the left side to the body; if the heart's 4 chambers fill 80 mL total equally, how much does one chamber hold? [\[Transportation in Animals & Plants\]](#)
- (A) 80 mL; the heart has four chambers
 - (B) 16 mL; the heart has four chambers
 - (C) 10 mL; the heart has four chambers
 - (D) 20 mL; the heart has two chambers
 - (E) 40 mL; the heart has four chambers
 - (F) 20 mL; the heart has four chambers
 - (G) 320 mL; the heart has four chambers
- 84.** A floor needs 0.45 litre of polish per square metre. After polishing 8 square metres, 1.4 litres are left. How much polish was there at the start? [\[Fractions & Decimals\]](#)
- (A) 9.4 litres
 - (B) 4.5 litres
 - (C) 3.95 litres
 - (D) 2.2 litres
 - (E) 4.6 litres
 - (F) 5.4 litres
 - (G) 5 litres
- 85.** A metal bridge is 500 m long and gains 1 cm of length for every 10 deg C rise; if the temperature climbs 40 deg C from morning to noon, how much longer does it get and what property explains it? [\[Heat\]](#)
- (A) 1 cm longer; solids expand when heated
 - (B) 4 cm shorter; solids expand when heated
 - (C) 4 cm longer; solids expand when heated
 - (D) 4 m longer; solids expand when heated
 - (E) 4 cm longer; metals are good insulators
 - (F) 0.4 cm longer; solids expand when heated
- 86.** Gold and copper in an alloy are in ratio 5:1 by weight. To change the ratio to 5:2, how much copper must be added to 600 g of alloy? [\[Comparing Quantities\]](#)
- (A) 100 g
 - (B) 50 g
 - (C) 150 g
 - (D) 80 g
 - (E) 120 g
 - (F) 200 g
 - (G) 60 g

- 87.** Root hairs absorb water; if one root hair takes in 0.2 mL daily and a root has 500 root hairs, how many litres of water does the root absorb in 5 days? [\[Transportation in Animals & Plants\]](#)
- (A) 0.05 litres; root hairs absorb water from soil
 - (B) 2.5 litres; root hairs absorb water from soil
 - (C) 0.5 litres; root hairs absorb water from soil
 - (D) 0.1 litres; root hairs absorb water from soil
 - (E) 5 litres; root hairs absorb water from soil
 - (F) 0.5 litres; stomata absorb water from soil
- 88.** If $\frac{2}{3}$ of a number equals 0.5 of 48, what is the number? [\[Fractions & Decimals\]](#)
- (A) 72
 - (B) 18
 - (C) 36
 - (D) 30
 - (E) 24
 - (F) 32
- 89.** Sitting near a campfire you feel warmth on your face without touching the flame or any moving air; name this transfer, and if you step twice as far so warmth drops to one quarter, how warm is it compared to before? [\[Heat\]](#)
- (A) One eighth as warm; radiation needs no medium to travel
 - (B) One quarter as warm; radiation needs a medium to travel
 - (C) Twice as warm; radiation needs no medium to travel
 - (D) One quarter as warm; radiation needs no medium to travel
 - (E) Half as warm; radiation needs no medium to travel
 - (F) One quarter as warm; conduction needs no medium to travel
 - (G) Four times as warm; radiation needs no medium to travel
 - (H) One quarter as warm; convection needs no medium to travel
- 90.** Ravi finished 0.875 of a job on Monday and 0.6 of the rest on Tuesday. What total percent of the whole job is finished? [\[Comparing Quantities\]](#)
- (A) 87.5%
 - (B) 52.5%
 - (C) 95%
 - (D) 94%
 - (E) 92.5%
 - (F) 90%
 - (G) 97.5%
- 91.** Double circulation means blood passes through the heart twice per full body trip; if the heart pumps 75 times a minute, how many beats occur in 8 minutes? [\[Transportation in Animals & Plants\]](#)
- (A) 480 beats; double circulation passes heart twice
 - (B) 4800 beats; double circulation passes heart twice
 - (C) 1200 beats; double circulation passes heart twice
 - (D) 150 beats; double circulation passes heart twice
 - (E) 600 beats; single circulation passes heart once
 - (F) 600 beats; double circulation passes heart twice

92. Each step of a staircase is 0.18 metre high. A child has climbed $\frac{2}{3}$ of a staircase that is 2.7 metres tall. How many steps has the child climbed? [\[Fractions & Decimals\]](#)

- (A) 6
- (B) 8
- (C) 12
- (D) 18
- (E) 10
- (F) 9

93. A cup of coffee cools from 80 deg C to 50 deg C in 15 minutes; if it cools at a steady rate, what is the cooling rate per minute and which transfer carries heat from the surface to the room air? [\[Heat\]](#)

- (A) 2 deg F per minute; convection carries heat to the room air
- (B) 30 deg C per minute; convection carries heat to the room air
- (C) 0.5 deg C per minute; convection carries heat to the room air
- (D) 5 deg C per minute; convection carries heat to the room air
- (E) 2 deg C per minute; radiation carries heat into the cup
- (F) 3 deg C per minute; convection carries heat to the room air
- (G) 2 deg C per minute; convection carries heat to the room air
- (H) 2 deg C per minute; conduction carries heat to the room air

94. A dealer allows a 10% discount and still gains 17% on cost. If the marked price is Rs 1300, find the cost price. [\[Comparing Quantities\]](#)

- (A) Rs 1300
- (B) Rs 1053
- (C) Rs 1080
- (D) Rs 1000
- (E) Rs 950
- (F) Rs 1170

95. Capillaries are very thin so materials exchange easily; if a capillary wall is just 0.001 mm and an artery wall is 1.2 mm, how many times thicker is the artery wall? [\[Transportation in Animals & Plants\]](#)

- (A) 120 times; capillaries have the thinnest walls
- (B) 1.2 times; capillaries have the thinnest walls
- (C) 12 times; capillaries have the thinnest walls
- (D) 12000 times; capillaries have the thinnest walls
- (E) 1200 times; capillaries have the thinnest walls
- (F) 1199 times; capillaries have the thinnest walls
- (G) 1200 times; arteries have the thinnest walls

96. Order from smallest to largest: 0.6, $\frac{5}{8}$, and $\frac{3}{5}$. Which value is the largest? [\[Fractions & Decimals\]](#)

- (A) 0.58
- (B) They are all equal
- (C) 0.6 and $\frac{5}{8}$ are equal
- (D) $\frac{5}{8}$
- (E) $\frac{3}{5}$
- (F) 0.65
- (G) 0.6
- (H) $\frac{3}{5}$ and 0.6 differ

97. A bimetallic strip of brass and iron bends when heated because brass expands more; if brass grows 6 units and iron grows 4 units for the same heat, what is the difference that causes the bend and why does it curve? [\[Heat\]](#)

- (A) 2 units difference; the strip bends toward the cooler side only
- (B) 2 units difference; the strip bends toward the more-expanding brass
- (C) 2 units difference; the strip bends toward the less-expanding iron
- (D) 24 units difference; the strip bends toward the less-expanding iron
- (E) 10 units difference; the strip bends toward the less-expanding iron
- (F) 1.5 units difference; the strip bends toward the less-expanding iron

98. A worker's daily wage rose from Rs 500 to Rs 575, then he worked 4% fewer days this month. By what percent did the daily wage rise, and is his per-day pay up or down? [\[Comparing Quantities\]](#)

- (A) 15%; down
- (B) 4%; up
- (C) 13%; up
- (D) 20%; up
- (E) 12%; up
- (F) 15%; daily pay up 15%
- (G) 75%; up

99. A plant in sunlight transpires 9 mL per hour but only 3 mL per hour in shade; if it gets 5 sunny hours and 4 shady hours, what is its total water loss for the day? [\[Transportation in Animals & Plants\]](#)

- (A) 57 mL; light increases transpiration rate
- (B) 57 mL; shade increases transpiration rate
- (C) 45 mL; light increases transpiration rate
- (D) 108 mL; light increases transpiration rate
- (E) 12 mL; light increases transpiration rate
- (F) 81 mL; light increases transpiration rate

100. A tailor cuts cloth into strips of $\frac{3}{8}$ metre each from a 7.5 metre roll. After cutting the maximum whole strips, how much cloth is wasted? [\[Fractions & Decimals\]](#)

- (A) 0.5 metres
- (B) 0 metres
- (C) 0.125 metres
- (D) 20 metres
- (E) 0.375 metres
- (F) $\frac{3}{8}$ metre
- (G) $\frac{1}{4}$ metre